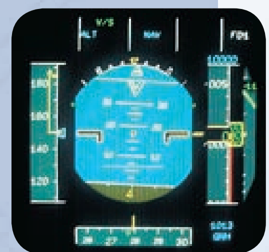
FLYING A MODERN AIRLINER

A modern jet is typically flown by a flight deck crew of two – a pilot, and a co-pilot who acts as navigator and flight engineer. The number of aircrew has been reduced through the use of computers and improvements in instrumentation and navigational equipment. In the latest airliners, color screens have mostly superseded dials for the display of data on altitude, airspeed, position, and other essentials.

Navigation has achieved a high degree of accuracy. It still partly relies on radio beacons that provide waypoint markers along airways, although these now operate on VHF. The co-pilot tunes in a radio compass to each successive beacon along the marked route. On-board inertial navigation systems (INS) will plot the aircraft's movements to within about a mile tolerance on a long-haul flight, if its position is set correctly at the start of the trip. Sophisticated airports have VOR (VHF omnidirectional range transmitters) with DME (distance measuring equipment), allowing the aircrew to track their position relative to their destination. In recent years, GPS satellite positioning has been added to the array of navigational improvements. Satellites also provide superb weather forecasting, a major contribution to safety and comfort.

For most of a flight, the pilot's main task is to supervise the work of the autopilot and other on-board computers. ILS (instrument landing systems) are now of high quality at many airports. Responding to signals from a transmitter on the runway, on-board ILS tells the pilot if he is too high or too low, to the left or right of the correct approach path. ILS linked to an autopilot can deliver automatic landing in poor visibility. But safety still depends on pilots' alertness, judgment, and flying skill.



DISPLAYS
The modern instruments system centres around multi-functional screens, including a navigation display (right), and primary flight display (below), which includes an artificial horizon.

A320 COCKPIT
The Airbus A320 was the first airliner with fly-by-wire controls and a "glass cockpit," with a small number of colored display screens replacing the dozens of potentially confusing dials and gauges found in earlier airliner cockpits.

The power of the world-shrinking jets to promote social and economic change was unquestionable.

As early as the 1960s, for example, the tranquil Mediterranean island of Majorca was abruptly transformed into a package-vacation destination through the building of an airport – its shores quickly becoming lined with high-rise hotels. The same fate befell the Portuguese Algarve, the Canary Islands, and some parts of Greece. By the 1990s, the falling cost of long-haul flights had extended mass tourism to Thailand and Goa, Sri Lanka and Gambia, transforming the lives of local inhabitants.

Tourism also grew into a dominant economic activity in the world's major cities: the centers of London, Paris, and New York became as much destinations to visit as places to work or live in. Nor was all non-business travel for pleasure: millions of Muslims were able to make the pilgrimage to Mecca thanks to aircraft.

Meanwhile freight carried in jet transports altered patterns of consumption. Soon no one was surprised to find fresh fish from an African lake in a British supermarket or fresh flowers from Mauritius decorating a restaurant table in Chicago.

While airliners carried on changing the world, after 1970 the world of airliners changed comparatively little. The last 30 years of the 20th century brought no further revolution in speed or size. But progress in engine design made aircraft quieter and more fuel efficient, and increased operating range – the Boeing 747-400 could fly almost 7,000 miles with more than 400 passengers